Computer Architecture- Quiz #5-C

Name: ________________________________ **Roll#:**_______________________

Section:_______

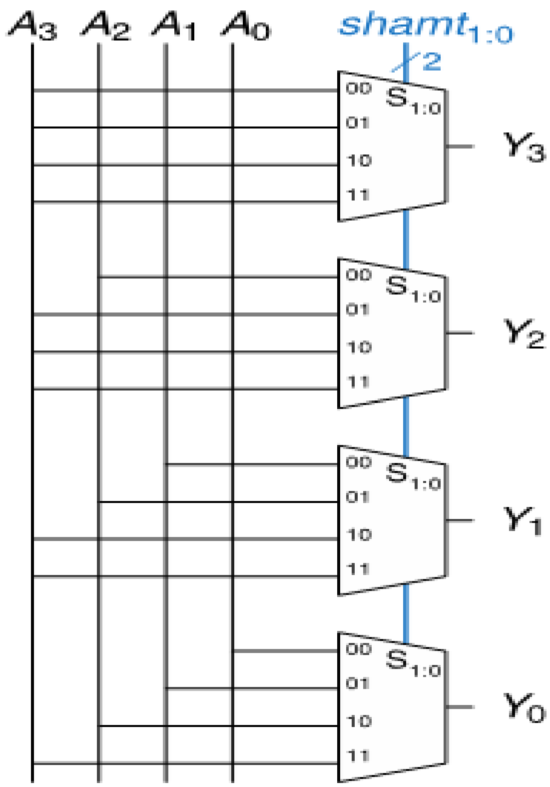
Total Time: 20 minutes Total Marks: 20 7th April, 2026

CLO 2: **Question 1** **Marks 15**

Propagation delay of a TTL based Full adder is approximately **10 ns**. Calculate delay for a **16-bit ripple carry adder** based on TTL based full adders? **(2)**
Consider **Carry-Lookahead Adder** and Write generic single Bit expressions for generate (G_i), propagate (P_i) signals and Carryout signals. **(3)** $\boldsymbol{G_i}$ = , $\boldsymbol{P_i}$ = $\boldsymbol{C_i}$ =
Calculate Column propagate $\boldsymbol{P_i}$ and generate signals $\boldsymbol{G_i}$ for each column if A =11001, B=10111
For a **4-bit carry look-ahead adder**, derive the expressions for the block propagate $\boldsymbol{P_{3:0}}$ and block generate $\boldsymbol{G_{3:0}}$ signals. Also, express the carry-out $\boldsymbol{C_3}$ in terms of these block propagate and generate signals. **(5)** $\boldsymbol{P_{3:0}}$ = $\boldsymbol{G_{3:0}}$ = C_3 =

CLO 2:	Question 2	Marks 5

Consider the given circuit. Evaluate its operation for input $A = 10$ and shift amount $shamt = 2$. Determine the final output $Y_{0:3}$ value in decimal form and identify the type of the circuit.



$Y_{0:3}$ = **Name:**

CLO 2:	Question 3	Marks 5

Design a 2 x 2 multiplier?